

Types of blockchains

Blockchain technology comes in various types, each designed for specific purposes. Here's an overview of the three main types.

Public blockchains: Public blockchains are open to anyone, making them entirely decentralised. These blockchains are maintained by a global network of nodes, and anyone can join, participate, and validate transactions. Public blockchains offer a high degree of transparency and security. They are ideal for applications where trust among participants is paramount, such as cryptocurrencies.

Private blockchains: Private blockchains, as the name suggests, restrict access. They are used primarily within organisations for internal purposes. Access to a private blockchain is controlled, typically by a central entity or group of entities. This results in faster transaction processing and greater privacy compared to public blockchains.

Consortium blockchains: Consortium blockchains strike a balance between public and private blockchains. They are governed by a pre-selected group of organisations or nodes. Consortium blockchains are semi-decentralised, combining the controlled access of private blockchains with the collaborative features of public blockchains. They are often used in industries where multiple organisations need to collaborate while maintaining some level of trust and privacy.

Use cases and examples of each type

These use cases explore the real-world impact of blockchain's versatility. Let's take a look at how they shape industries and redefine trust.

Public blockchain use cases

- **Bitcoin**, the pioneer of cryptocurrencies, operates on a public blockchain. It allows anyone

to participate in the network, verify transactions, and earn rewards (in the form of new Bitcoins) for contributing computational power. Bitcoin's primary use case is as a digital currency, enabling peer-to-peer transactions without the need for intermediaries like banks.

- **Ethereum** is another well-known public blockchain, but its use cases go beyond digital currency. Ethereum introduced smart contracts, which are self-executing contracts with the terms directly written into code. These contracts run on the Ethereum blockchain and enable automated and secure execution of agreements. For example, digital artists can create a smart contract that ensures they receive a percentage every time their artwork is resold, empowering creators in the digital realm.

Private blockchain use case

- **Hyperledger fabric** is a prominent example of a private blockchain framework. It's designed for use within organisations and consortiums. Hyperledger fabric offers enhanced privacy and scalability while maintaining control over the network. Use cases for this framework include supply chain management, where companies need to securely share information about the movement of goods without exposing sensitive data to the public.

Consortium blockchain use case

- **R3 Corda** is a consortium blockchain platform designed for businesses and financial institutions. It allows multiple parties to collaborate while preserving confidentiality. Corda is used in scenarios like trade finance, where banks, corporations, and regulators need to exchange information and assets securely. For instance, it can facilitate the transfer of ownership of goods as they move through a supply chain while ensuring

that only authorised parties have access to relevant data.

Importance of open source for blockchain

Open source and blockchain are kindred spirits, both embodying transparency, collaboration, and community-driven innovation. Open source means the source code is freely available for inspection, modification, and enhancement by anyone. In the blockchain realm, this philosophy is pivotal. Consider Ethereum, a decentralised platform that allows developers to build and deploy smart contracts. Its open source nature ensures that the platform evolves based on collective intelligence, promoting security and adaptability.

Open source blockchain use cases

Open source blockchain technology has unlocked a treasure trove of real-world applications. Let's dive into some examples.

Supply chain authentication with VeChain: VeChain, an open source blockchain platform, partners with companies to verify the authenticity and origin of products. For instance, luxury brand products can be traced from creation to the store, assuring buyers that they're receiving genuine items.

Decentralised identity with Sovrin: Sovrin, an open source blockchain network, addresses the challenges of managing digital identities. It enables individuals to control their personal information securely, reducing the risks of data breaches and identity theft.

Energy trading with Powerledger: Powerledger leverages blockchain to enable peer-to-peer energy trading. Home owners with solar panels can sell excess energy to their neighbours, promoting sustainable energy usage.

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